

CCNA 200-301 Notes

Chapter 04 - Network Media & Ethernet Cables

Topics Covered

- Network Media Types
- Twisted Pair
- Straight Through
- Crossover
- Rollover
- UTP vs STP
- UTP Categories
- 100m Rule

Key Notes

- Coaxial is an old cable technology.
- Twisted Pair is the most common LAN cable.
- Fiber Optic is the fastest and supports long distances.
- Twisted Pair has 8 copper wires (4 twisted pairs).
- Twisting reduces EMI and Crosstalk.
- Straight Through connects different devices.
- Crossover connects similar devices.
- Rollover is used for console connections.

UTP vs STP

UTP	STP
No Shield	Shielded
Cheaper	More Expensive
Easy Installation	Better EMI Protection

Quick Review

- Cat5 → 100 Mbps
- Cat5e → 1 Gbps
- Cat6 → Up to 10 Gbps (short distance)
- Maximum UTP Length = 100 meters

Important Scenario ☐

If the cable distance is more than 100 meters (e.g. 150m or 200m), place a network device in the middle to regenerate the signal.

Traditionally this can be a Repeater. In modern networks, a Switch is commonly used instead of a Hub.

Exam Tips ☐

- Straight = Different Devices
- Crossover = Similar Devices
- Fiber = Fastest
- Twisting reduces interference, not speed
- After 100m, use a Switch (or Repeater) to extend the connection

30-Second Revision

- Coaxial → Old
- Twisted Pair → Most Common
- Fiber → Fastest
- Straight → Different
- Crossover → Similar
- Rollover → Console
- Max Length → 100m
- 150m+ → Add Switch / Repeater

Cisco Commands

No Cisco IOS commands in this lesson.